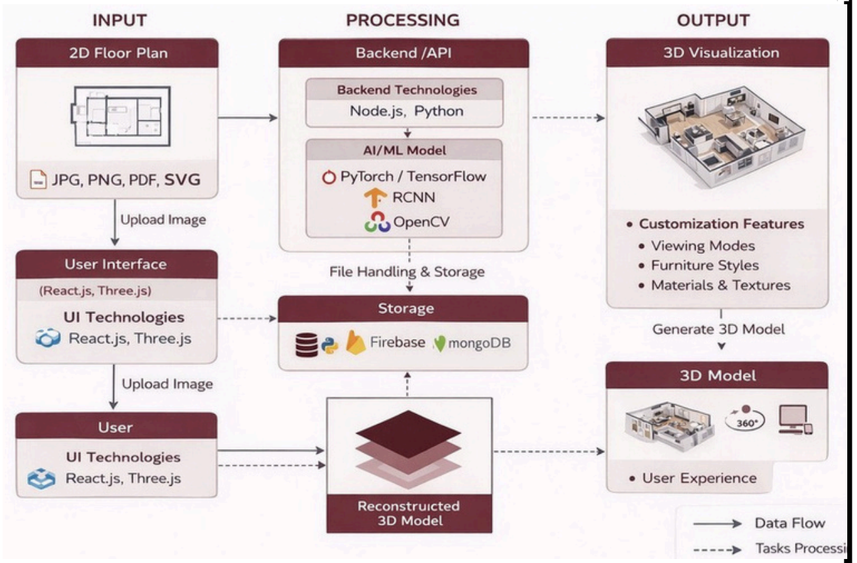
PLANORA: A Blend of Architecture and Vision

Problem Statement:

Converting 2Dfloor plansinto 3Dmodels is time-consuming and require expertise in complex CAD tools. Existing solutions lack automation and real-time visualization making it difficult for users to understand spatial layouts.

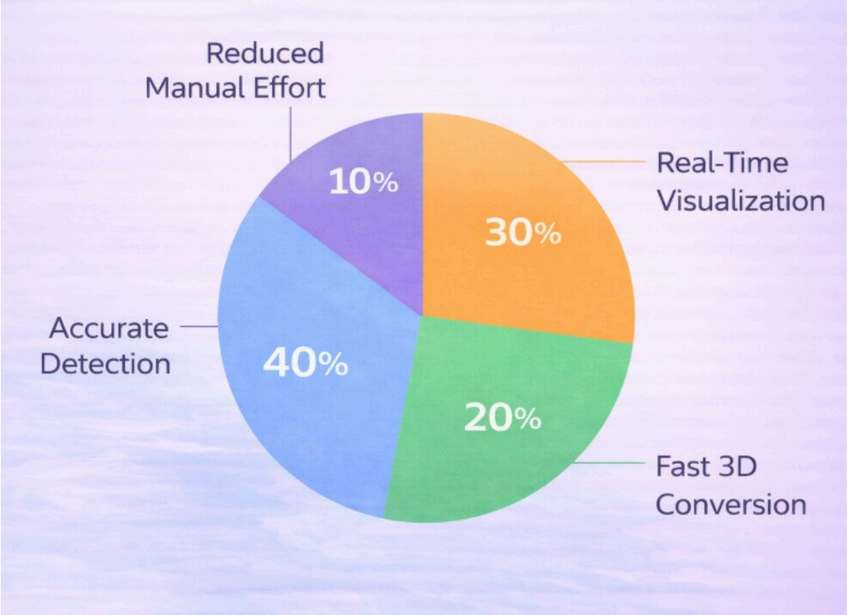
Block Diagram:



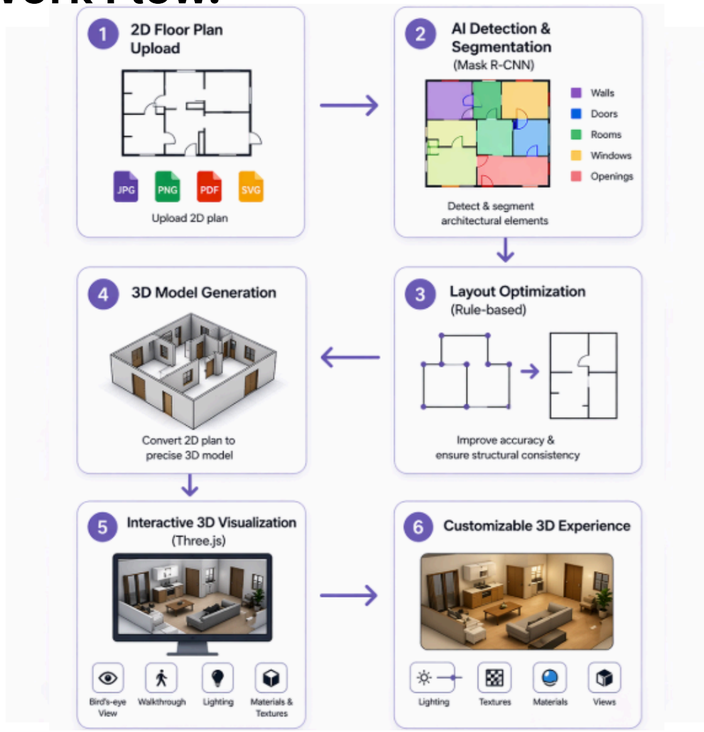
Objectives:

•To To enable users to upload 2D floor plans in JPG, PNG, PDF, or SVG formats.
•To convert 2D layouts into accurate 3D models using Al.
•To provide customizable rendering styles, lighting, and material options.
•To allow exploration through multiple viewing angles, including bird's-eye and walkthrough modes.

Results:



Work Flow:



References:

- He et al., Mask R-CNN, ICCV 2017
- Liu et al., Deep Floor Plan Recognition using CNNs, 2017

Conclusion:

Successfully developed a system to convert 2D floor plans into 3D models using AI. It enables real-time visualization, simplifying architectural design.